Sensitivity Analysis of the Relationship Between Chronic Obstructive Pulmonary Disease (COPD) and Psychological Distress

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Abstract
Sensitivity analysis confirmed or validated the significant variables (psychological distress, age, smoking and general health) in determining the odds of having chronic obstructive pulmonary disease (COPD), found in the published logistic regression model (Kham-Ai, Heaton, and Li, 2023, p.92). These variables retained their significance, if unmeasured confounding around psychological distress or the small sample size was ignored. Sensitivity analysis indicated that general health could be estimated with drastic reductions in uncertainty, further validating the robustness of the published results.

Keywords:
Farmers; COPD; Psychological Distress; Covariates; Simple Logistic Regression; Bayesian Logistic Regression; Firth Penalized Logistic Regression; Sensitivity Analysis; Aleatory Uncertainty; Epistemic Uncertainty

0.1 Introduction

Published research, that investigated the association between COPD and psychological distress in alleged farmers indicated that ***psychological distress***, ***age***, ***smoking***, and ***general health*** were statistically significant variables in a *simple logistic regression (SLR)* model (Kham-Ai, Heaton, and Li 2023).

All models almost certainly possess ***aleatory*** and/or ***epistemic*** uncertainty. Marino et al. (2008) defined *epistemic uncertainty* as model output instability stemming from experimental failures (survey execution problems), while *aleatory uncertainty* was defined as model output instability driven by stochastic or random components of the model. *Epistemic uncertainty* and *aleatory uncertainty* can become conflated in the model output, hence there is a need to differentiate these concepts in the implementation of a sensitivity analysis (Marino et al. 2008). The sample size ($n = 239$) utilized by the authors, underscored the high likelihood of both *aleatory* and *epistemic* uncertainty. The simple logistic regression (SLR) model (`COPD ~ Psychological distress + Covariates`) was assessed for *aleatory uncertainty* by modifying model components, such as the addition of a regularization component in the Firth Penalized Logistic Regression (FPLR) and, the use of an informative prior component (Bayes theorem) in the Bayesian Logistic Regression (BLR). *Epistemic uncertainty* was assessed by modifications of the model inputs through data simulations (expanding the sample size), bootstrapping, leave one out cross validation (LOOCV), outliers-excised data, local changes on variables like general health and psychological distress.

Sensitivity analysis was done through visual (`ggplot` graphs) comparison of the coefficients or Odds Ratios (ORs), and ORs precision (via observation of the confidence intervals) across different scenarios, to determine potential model modifications or input(s) variations, needed to improve estimation precision, or simply to validate or reject any assertions made by the published authors.

0.2 Methods

The methodology of analysis is provided here in a clear format. The methodology progresses from *data sources*, *variables or measures*, *data preparation* and a *short description of the methods used*.

0.2.1 Data Source(s)

This sensitivity analysis utilized 2017 data from the **Centers for Disease Control (CDC) Behavioral Risk Factor Surveillance System (BRFSS)**. The BRFSS is an annual telephonic health survey system established in 1984, that collects health data from more than $> 400,000$ US residents (aged 18 and above) across all 50 states, related to their *health-related risk behaviors*, *healthcare access*, *chronic health conditions*, and *use of preventive services*.

0.2.2 Variables or Measures

The binary outcome variable was ***COPD***. Alleged farmers that answered "Yes" to having 1 of 3 COPD symptoms (Cough, phlegm or breathing issues) in the past 3 months, were considered to be COPD Positive, otherwise

they were considered COPD negative (visit the Appendix for details).

The main predictor was ***psychological distress***. An answer of, "A little of the time", "Some of the time", "Most of the time" or "All of the time", to the query *"Within the last 30 days how often have you felt this kind of stress?"*, were classified as distressed, otherwise, not distressed.

The covariates or confounders in the model were *smoking status, age, gender, race, income, and general health*. Descriptions of the covariates used in the sensitivity analysis can also be found by visiting the Appendix.

0.2.3 Data Preparation

Published sample: The initial dataset consisted of 450,016 subjects with a sample space of 358 variables. The sample space was reduced to 8 variables. The included variables were COPD, Psychological Distress, Smoking, Age, Gender, Race, Incomes, and General Health Status. (Kham-Ai, Heaton, and Li 2023).

The BRFSS data had no variable for occupations. To extract alleged farmers, individuals who reported they did farm or ranch work for the question "What type of physical activity or exercise did you spend the most time doing during the past month" and the question "What other type of physical activity gave you the next most exercise during the past month?" ($N = 2994$).

Alleged farmers were filtered to obtain subjects that answered questions on 1 of 3 COPD symptoms ($n = 318$). Alleged farmers with COPD answered "yes" to 1 of 3 questions; During the past 3 months, did they have a cough on most days?, did they cough up phlegm [FLEM] or mucus on most days?, Did they have shortness of Breath (SOB) either when hurrying on level ground or when walking up a slight hill or stairs?

Alleged farmers with a COPD response were further filtered to obtain subjects that provided a response to the social or psychological distress prompt ($n = 240$). COPD subjects who responded to, Within the last 30 days, how often have you felt this kind of stress?

An arbitrary step was used to filter out a subject with an invalid response to COPD symptoms, which arrived at the same sample size of the published paper($n = 239$).

Completed Cases: Analysis results can be affected by including invalid responses across the sample space. The published sample was filtered to remove all invalid or unusable responses (Refused, Unknown, etc.) across the entire sample space ($n = 199$). Noise from model inputs can be prevented by using completed cases in the sensitivity analysis.

Simulated Sets: As indicated earlier, the small sample size ($n = 239$) used in the publication was certainly going to lead to uncertainty in the model coefficients. Increasing knowledge about the sample space entailed the use of simulated data by performing the imputation of missing values. Imputation of missing values was performed by using an advanced technique known as multiple imputation (Zhang 2016).

Data with all the screened alleged farmers ($N = 2994$) was obtained, with all the missing or invalid responses. The data was prepared for imputation by converting invalid responses into missing responses (Zhang 2016). Multiple imputation was used to replace missing values with their best approximations. Multiple datasets (generally 5 datasets) were generated using the imputation procedure and pooled into a list, for further analysis.

0.2.4 Study Replication

Performing the sensitivity analysis required replicating the logistic regression model performed by Kham-Ai, Heaton, and Li (2023). Sensitivity analysis of a model, requires a valid model. The ***simple logistic regression (SLR)*** utilized in the published paper was checked for adherence to major assumptions of a logistic model.

The assumption of a binary variable was checked by investigating the number of levels in the COPD variable, and it was confirmed the outcome was binary.

The assumption of linearity of the relationships between the *log odds* of the outcome and predictors is required because the logistic regression has a linear form ($y = mx + c$). A scatter plot of *log odds* against the mean(s) of numeric predictor(s), detected no violation of the linearity assumption.

Logistic regression models are more robust and reliable, with no multicolinearity between predictors. Using the *variance inflation factor* as a metric, indicated no multicolinearity between predictors.

Logistic regression models are reliable, with no extreme outliers in the data. Using *Cook's distance* (a measure of influential data points) as a metric, indicated no extreme outliers capable of violating the outlier assumption.

The assumption of independent observations was satisfied. With no violations of assumptions, the SLR model from Kham-Ai, Heaton, and Li (2023) was implemented (See **Table 1** for the model).

0.2.5 Sensitivity Analysis Techniques

Sensitivity analysis explores *epistemic* and *aleatory* uncertainty. The methods implemented are outlined under the appropriate type of uncertainty.

Global Aleatory uncertainty

Firth Penalized Logistic Regression (FPLR): SLR is a viable tool for understanding binary outcomes in large enough datasets, and often performs best in the absence of class imbalances. Model stability is often necessary for the consistency of model outputs. FPLR tweaks model components by adding a penalty term (regulariza-

tion parameter) known as *Jeffreys' invariant prior* which is a *Fisher information matrix* intended to capture or measure the amount of information carried by data about the parameters. The likelihood function is multiplied or added to jeffreys' penalty term on a non-log and log scale respectively. Theoretical conceptualization below:

$$\log p(\theta \mid X) = \ell(\theta \mid X) + \frac{1}{2}\log|\mathcal{J}(\theta)| + C \tag{1}$$

$$\ell(\theta \mid X) = \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{2}$$

$\log p(\theta \mid X)$: The Log-Posterior Density is the natural Log of the probability distribution (PD) of the parameters (θ) given the observed data (X).

$\ell(\theta \mid X)$: The Log-Likelihood Function is used to determine the optimum parameters (θ) that explain the observed data (X).

$\mathcal{J}(\theta)$: The Expected (or Observed) Fisher Information Matrix measures the amount of information that the observed data (X) carries about the unknown parameters (θ). The determinant ensures regularization that can lead to 0 shrinkage of parameters.

C: The Normalization Constant is a constant independent of the parameters (θ), ensuring that the total probability integrates to 1.

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically, by comparing FPLR vs SLR outputs.

Bayesian Logistic Regression (BLR): BLR is beneficial when prior information on the model parameters are known. BLR permits the investigation of model stability by approximation of parameters as conjugate Gaussian or normal distributions (random variables) given by $p(\beta|y, X) = \mathcal{N}(\beta \mid \mu_N, \Sigma_N)$, compared to fixed constants in simple logistic regression. Bayes theorem is theoretically modeled below:

$$P(Hypothesis(H)|Evidence(E)) = \frac{P(E|H)P(H)}{P(E)} \tag{3}$$

$$P(\beta \mid X, Y) = \frac{P(Y \mid X, \beta)P(\beta)}{P(Y)} \tag{4}$$

The BLR model is conceptualized as such:

$$p(\beta \mid X, Y) = p(\beta)p(Y|X, \beta) \tag{5}$$

$$p(Y|X, \beta) = L(\theta \mid X) = \prod_{i}^{N} \sigma(X_i\beta)^{y_i}(1-\sigma(X_i\beta))^{1-y_i} \tag{6}$$

$$\begin{aligned} p(\beta) &= \mathcal{N}(\beta \mid \mu_0, \Sigma_0) \\ &= (2\pi)^{-\frac{D}{2}}|\Sigma_0|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\beta - \mu_0)^T\Sigma_0^{-1}(\beta - \mu_0)\right) \end{aligned} \tag{7}$$

where Σ_0 is the prior covariance matrix and μ_0 is the prior mean.

$$p(\beta|Y, X) \propto p(Y|X, \beta) * p(\beta) \tag{8}$$

$$p(\beta|Y, X) = \mathcal{N}(\beta \mid \mu_N, \Sigma_N) \tag{9}$$

Parameters obtained from BLR are in the form of normal distributions (random variables), as in equation 9. *Maximum Aposteriori Estimation (MAP)* is used to guess the best value of the parameter, which is often the mode of the posterior distributions.

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically, by comparing BLR vs SLR outputs.

Global Epistemic uncertainty

Bootstrapped SLR: The published sample ($n = 239$) was bootstrapped or resampled a 1000 times with replacement, in order to understand the impact of random data configurations on the Odds Ratios (ORs) and Confidence Intervals (CIs). If the sample were a very close approximation of the population, how do the parameters and certainty of estimation differ?

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically, by comparing Bootstrapped SLR vs other outputs.

Leave One Out Cross-Validation SLR: The published sample ($n = 239$) was used to fit a logistic regression n times, while leaving out a single observation in every iteration, and final estimate and confidence interval was pooled from all iterations. If the sample were a very a remote approximation of the population, how do the parameters and certainty of estimation differ?

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically, by comparing LOOCV SLR vs other outputs.

Outliers-excised SLR: The presence of outliers is often considered misinformation. In rare circumstances, outliers can be overriding factors, as in the field of genetics. The SLR model was fitted to data without extreme outliers; extreme outliers were defined using Cook's distance which is a measure of how influential a data point is on model coefficients.

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically, by comparing the Outlier-excised SLR vs SLR.

Sample Size Varied SLRs: Manipulating knowledge by using actual data was achieved via data filtering to get completed cases or using simulated cases , as explained in the Data Preparation section. SLR was performed on the simulated sets and completed cases.

It must be stated that, the simulated datasets did not account for the veracity of knowledge but rather helped in ensuring complete information that closely approximated distributions based on known information. Simulated data still produced a good foundation for knowing the model outputs if more knowledge was obtained around the sample space, irrespective of the truth.

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically (See **Table 2**), by comparing the SLR models with different sample sizes.

Local Epistemic uncertainty

Categorical Variable Collapse: The *general health* covariate was shown to have been unreliably estimated given its very wide confidence interval when SLR was applied to the published sample ($n = 239$), and the unusable confidence interval when the SLR was applied to the completed cases dataset ($n = 199$). The variable was collapsed into a binary variable denoted as "Good" versus "Poor" *general health* (See the Appendix for a description of the original and modified variable).

The OR and CI for this variable alone was observed to see any improvement or deterioration in the OR estimation precision.

Redefining Psychological Distress: The main predictor was defined as psychological distress by the authors and given some familiarity with the dataset, it seemed possible to sensibly (with good reason) modify the distribution of this variable by ensuring that alleged farmers marked as being in psychological distress did not report having 0 days of bad mental health.

Mental health days was a numeric measure from the query, "Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?". Alleged farmers originally classified as psychologically distressed and reported greater than 0 days of poor mental health were classified as psychologically distressed, otherwise, not psychological distressed.

The incorporation of mental health days in coding for psychological distress also addressed some concerns of unmeasured confounding.

Odds ratio (OR) and OR estimation precision (confidence or credible interval(s)) were evaluated graphically (See **Table 2**), by comparing the SLR models with different approaches to the classification of psychological distress.

0.3 Results

The results for this paper are provided in the subsections below.

0.3.1 Study Replication

The study replication indicated that psychological distress, general health, smoking and age were statistically significant determinants of having COPD among farmers. The replicated results provided *qualitative* and near precise *quantitative* validation of the results from the published study being analyzed. The replication results are provided in **Table 1**.

0.3.2 Firth Penalized Logistic Regression

The SLR model was slightly sensitive to additional components (a regularization component known as Jeffrey's invariant prior), indicated by the shrinkage of confidence or credible intervals, and the subtle deviations in point estimates (ORs) with FPLR. The SLR outputs appeared robust enough to withstand model modifications, evidenced by retention of the same significant variables in the FPLR.

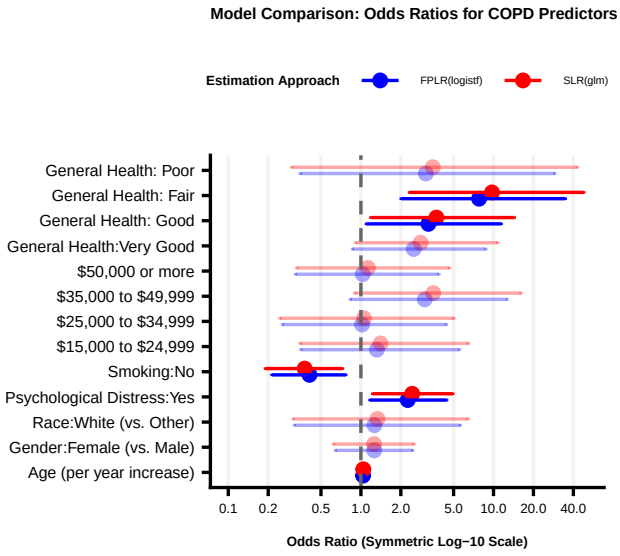


Figure 1: The SLR model was slightly sensitive to additional components (a special geometric prior regularization component), evidenced by the mild differences in point estimates and mild shrinkage of the confidence or credible intervals. The SLR outputs appeared robust enough to withstand model modifications, evidenced by retention of the same significant variables in the FPLR. Solid intervals are statistically significant at p <0.05 (95% CI excludes 1). Faded intervals are non-significant.

0.3.3 Bayesian Logistic Regression

The SLR model was sensitive to additional component(s) (a quadratic regularization component for a zero-mean specified multivariate normal prior distribution on the

Variable	Level	Overall	No COPD	COPD	P-Value
Overall cohort (n)		239	165	74	—
Age[a]		60.42 (13.42)	59.17 (13.94)	63.20 (11.81)	0.003**
General Health Status[b]	Excellent	37 (15.5%)	33 (20.0%)	4 (5.4%)	0.003**
	Very Good	90 (37.7%)	65 (39.4%)	25 (33.8%)	—
	Good	81 (33.9%)	53 (32.1%)	28 (37.8%)	—
	Fair	25 (10.5%)	11 (6.7%)	14 (18.9%)	—
	Poor	4 (1.7%)	2 (1.2%)	2 (2.7%)	—
Smoking[b]	Yes	104 (43.5%)	60 (36.4%)	44 (59.5%)	0.004**
	No	133 (55.6%)	103 (62.4%)	30 (40.5%)	—
Psychological distress[b]	Yes	130 (54.4%)	99 (60.0%)	31 (41.9%)	0.012*
	No	109 (45.6%)	66 (40.0%)	43 (58.1%)	—
Gender[b]	Male	168 (70.3%)	117 (70.9%)	51 (68.9%)	0.52
	Female	71 (29.7%)	48 (29.1%)	23 (31.1%)	—
Incomes[b]	< $15,000	19 (7.9%)	15 (9.1%)	4 (5.4%)	0.083.
	$15,000 to $24,999	33 (13.8%)	23 (13.9%)	10 (13.5%)	—
	$25,000 to $34,999	26 (10.9%)	17 (10.3%)	9 (12.2%)	—
	$35,000 to $49,999	31 (13.0%)	17 (10.3%)	14 (18.9%)	—
	$50,000 or more	96 (40.2%)	75 (45.5%)	21 (28.4%)	—
Race[b]	Other	13 (5.4%)	8 (4.8%)	5 (6.8%)	0.70
	White	226 (94.6%)	157 (95.2%)	69 (93.2%)	—

[a] Continuous variables are presented as Mean (Standard Deviation SD).
[b] Categorical variables are presented as Frequency (Percentage %).

Table 1: Study Replication: Baseline Demographics and Clinical Characteristics Stratified by COPD Status.

coefficients), indicated by the shrinkage of confidence or credible intervals, and the difference in point estimates (ORs) for the ***general health*** and ***income*** variables. The SLR outputs appeared robust enough to withstand model modifications, evidenced by retention of the same significant variables in the BLR.

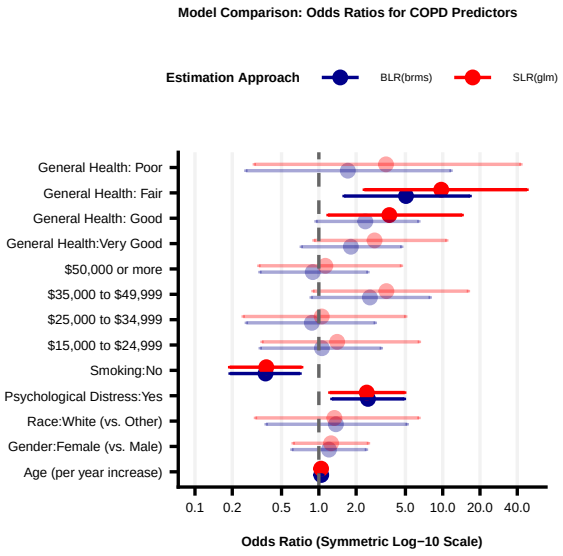


Figure 2: The SLR model was sensitive to additional components (a quadratic regularization component), evidenced by the noticeable difference(s) in point estimates for the general health and income variables. The SLR outputs appeared robust enough to withstand model modifications, evidenced by retention of the same significant variables in the FPLR. Solid intervals are statistically significant at $p < 0.05$ (95% CI excludes 1). Faded intervals are non-significant.

0.3.4 Bootstrapping Simple Logistic Regression

The SLR model was sensitive to utilizing a sample that was NOT a close approximation of the population. Variables like ***income*** and ***general health*** showed increased estimation imprecision. The SLR outputs appeared robust enough to withstand some lack of knowledge about the population from which the sample was drawn.

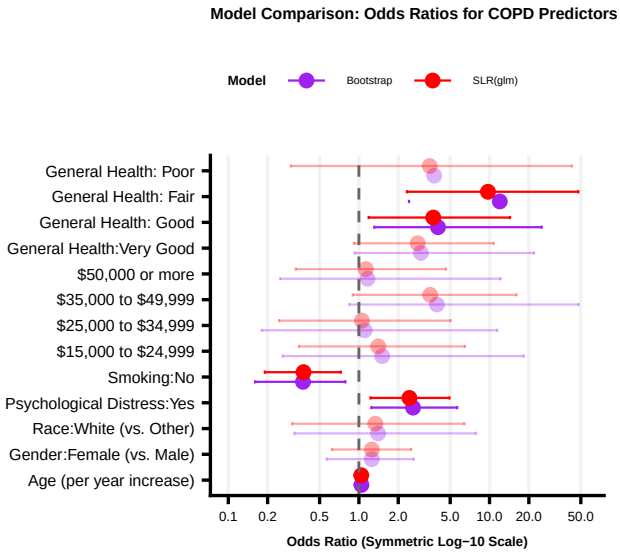


Figure 3: Variables like income and general health showed increased estimation imprecision but no drastic differences in point estimates. The SLR outputs appeared robust enough to withstand some lack of knowledge about the population from which the sample was drawn. Solid intervals represent statistical significance at $p < 0.05$ / 95% Credible Interval excludes 1. Faded intervals represent non-significant findings.

0.3.5 Cross Validating Simple Logistic Regression

The SLR model was insensitive to utilizing a sample that was a close approximation of the population. Point estimates were similar in leave one out cross validation (LOOCV) SLR and SLR. Estimation precision drastically improved in leave one out cross validation (LOOCV) SLR, relative to SLR.

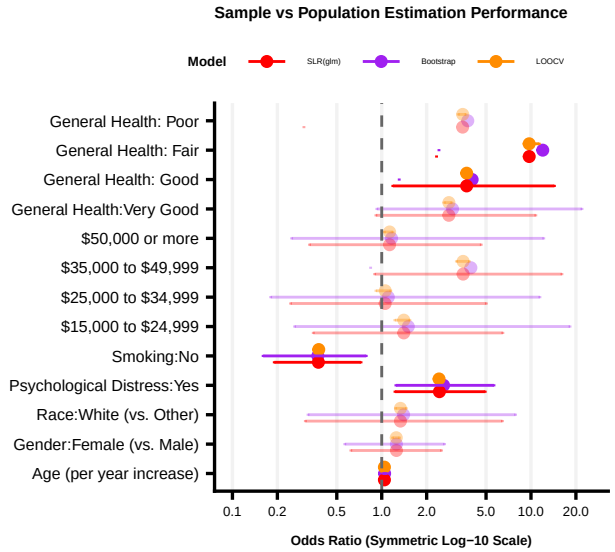


Figure 4: The SLR model was insensitive to utilizing a sample that was a close approximation of the population. There was no difference in the point estimates of the LOOCV SLR model vs the SLR model, and there was actually an improvement in estimation precision (shrinkage of confidence intervals) for the LOOCV SLR model. The SLR outputs appeared robust enough to withstand increased knowledge around the sample space through LOOCV, evidenced by retention of the same significant variables in the LOOCV SLR. Solid intervals represent statistical significance at $p < 0.05$ (95% CI excludes 1). Faded intervals represent non-significant findings.

0.3.6 Simple Logistic Regression Sensitivity to Outliers

The SLR model was sensitive to the presence of outliers within the sample, indicated by the noticeable difference(s) in point estimates (ORs) for the ***general health*** variable. The presence of outliers had global effects but these effects were more pronounced in the general health variable. SLR outputs appeared robust enough to withstand the presence of outliers, evidenced by retention of the same significant variables in the SLR model without outliers.

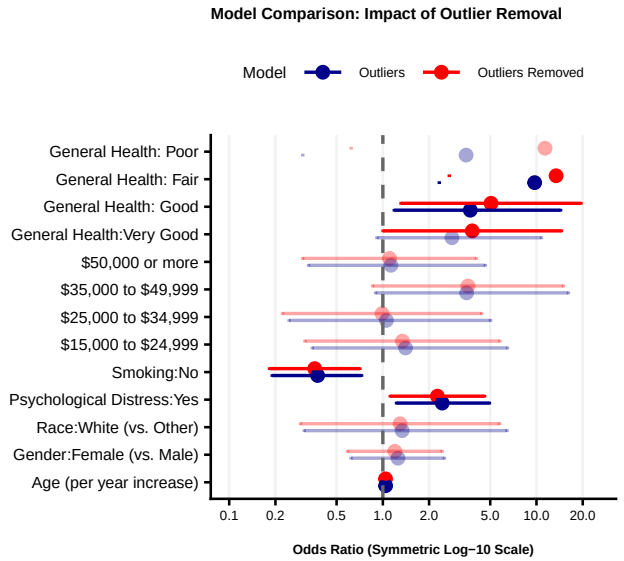


Figure 5: The SLR model was sensitive to the presence of outliers within the sample, indicated by the noticeable difference(s) in point estimates (ORs) for the general health variable. The presence of outliers had global effects but these effects were more pronounced in the general health variable. SLR outputs appeared robust enough to withstand the presence of outliers, evidenced by retention of the same significant variables in the SLR model without outliers. Solid intervals represent statistical significance at $p < 0.05$ (95% CI excludes 1). Faded intervals represent non-significant findings.

0.3.7 Knowledge Pertubations By Sample Size Variation

The baseline or reference simple logistic regression (SLR) on the published sample, recovered ***psychological distress***, ***smoking***, ***age***, and ***general health*** (depending on which level of the variable was reported) as statistically significant variables ($\alpha = 0.05$) in determining the odds of having ***COPD***.

Simple logistic regression (SLR) on a sample of completed cases ($n = 199$) around the sample space, indicated model sensitivity to disparities in knowledge about the sample space. The SLR model performed on the sample of completed cases, failed to retrieve ***smoking*** as a significant variable ($\alpha = 0.05$). SLR results for completed cases, indicated global increases in estimation imprecision (bloated confidence intervals).

Pooled Simple logistic regression (PSLR) on 5 samples of total cases ($n = 2994$) around the sample space, indicated model sensitivity to disparities in knowledge about the sample space. The SLR model performed on the sample of total cases, failed to retrieve ***psychological distress*** and ***age*** as significant variable(s) ($\alpha = 0.05$); ***race*** was retrieved as a significant variable but was not considered significant in the baseline or reference SLR model. The only baseline significant variable retained in the SLR model performed on the sample of total cases

was ***smoking*** ($\alpha = 0.05$). SLR results for total cases, indicated global increases in estimation precision (shrunk confidence intervals) for most variables except gender.

Results showing the comparison of SLR model performance at different knowledge thresholds about the sample space, were provided in **Table 2**.

0.3.8 Local Sensitivity Analysis on General Health

The SLR model output for general health, indicated local sensitivity to collapsing the levels found within the variable. Collapsing the variable from 5 levels to 2 levels that more clearly defined the groupings of subjects (alleged farmers), indicated local sensitivity of the model outcome for this variable (See the Appendix for a description of the original and modified variable).

Collapsing the general health variable into 2 levels ("Good" vs "Poor"), led to a drastic improvement in estimation precision, evidenced by shrinkage of the confidence interval for the general health variable.

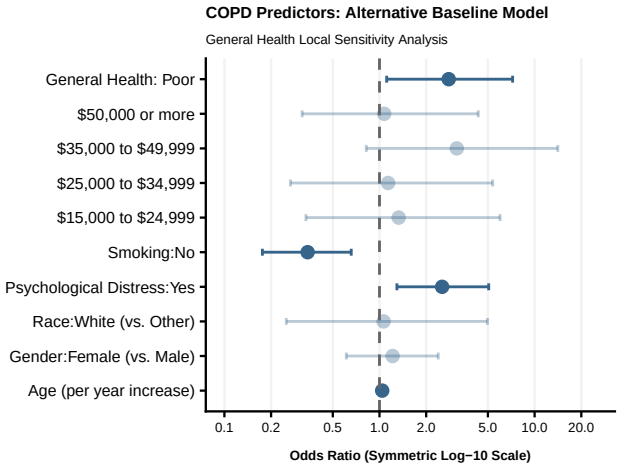


Figure 6: Collapsed levels for the general health variable, led to reduced uncertainty (improved estimation precision) for the ceofficient. The SLR outputs appeared robust enough to withstand, local changes to the general health variable, evidenced by retention of the same significant variables in the SLR model with the modified general health variable. Solid intervals represent statistical significance at $p < 0.05$ (95% CI excludes 1). Faded intervals are non-significant.

0.3.9 Local Sensitivity Analysis on Psychological Distress

The SLR model output for ***psychological distress***, indicated local sensitivity to unmeasured confounding around the variable. Utilizing the number of poor mental health days variable as a means to confirm psychological distress, indicated local sensitivity of the model outcome for this variable, evidenced by the difference in the point estimates for the SLR model(s) with alternate definitions of psychological distress.

The SLR model outcome for psychological distress was not robust enough to withstand unmeasured confounding around the variable, but the SLR outputs for other significant variables appeared robust enough to withstand the same unmeasured confounding.

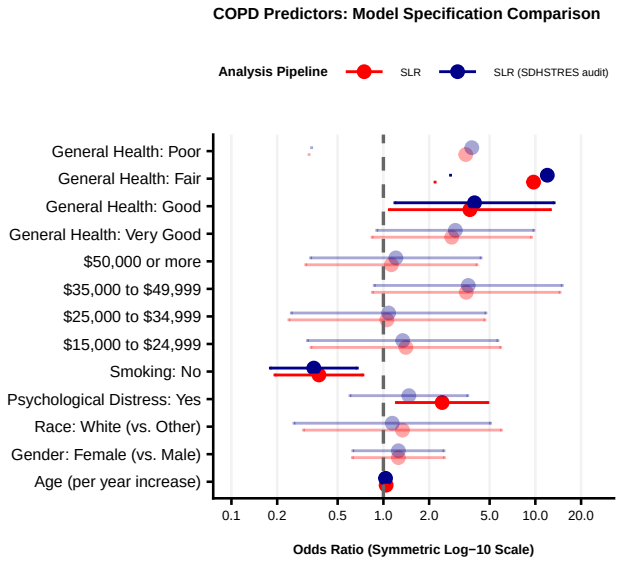


Figure 7: Redefining psychological distress, rendered the output for the variable insignificant, but the other significant variables had model outputs that were robust enough to be insensitive towards unmeasured confounding around psychological distress. Solid intervals represent statistical significance at $p < 0.05$ (95% CI excludes 1). Faded intervals represent non-significant findings.

0.4 Discussions

The results are discussed below, in the corresponding subsections matching the results.

0.4.1 Firth Penalized Logistic Regression (FPLR)

The FPLR model as a *global aleatory sensitivity analysis tool*, provided an opportunity to investigate the stability of the simple logistic regression (SLR) model. The FPLR model was derived by adding a special geometric regularization term or component (dubbed Jeffrey's invariant prior) to the baseline log likelihood function of the SLR (Visit the Firth Penalized Logistic Regression part in the methods section).

Jeffrey's invariant prior is a unique regularization component that does not look or behave like other well known regularization components, such as the ridge ($L2$) or lasso ($L1$) regularization component(s). Regularization components are thought of as global sensitivity analysis tools because they have global effects on all coefficients.

The FPLR model coefficients differed from the baseline SLR model, indicative of the SLR model sensitivity

Variable	Completed True Cases (N=199)		Effective Sample (N=239)		Pooled Simulations (N=2994)	
	aOR (95% CI)	P-Value	aOR (95% CI)	P-Value	aOR (95% CI)	P-Value
Psychological Distress:Yes	2.60 [1.21, 5.78]	0.016*	2.43 [1.22, 4.93]	0.012*	1.44 [0.88, 2.37]	0.14
Smoking:No	0.49 [0.23, 1.01]	0.055	0.37 [0.19, 0.72]	0.004**	0.40 [0.25, 0.64]	0.0001***
General Health:Poor[a]	0.000 [—, Inf]	0.98	3.48 [0.30, 42.78]	0.30	3.18 [0.69, 14.59]	0.13
Age	1.04 [1.01, 1.07]	0.009**	1.04 [1.01, 1.07]	0.003**	1.01 [0.99, 1.03]	0.18
Race:White	1.26 [0.25, 7.56]	0.77	1.33 [0.30, 6.43]	0.70	0.30 [0.095, 0.95]	0.04*
Gender:Female	0.81 [0.35, 1.78]	0.60	1.25 [0.62, 2.51]	0.52	1.62 [0.78, 3.36]	0.19

[a] Note: Low statistical estimation precision variable primed for Local Sensitivity Analysis (LSA).

Table 2: Sensitivity Analysis Logistic Regression: Adjusted Odds Ratios Across Completed True Cases, Effective, and Simulated Samples. In completed cases (n=199), smoking was not recovered as significant. In total cases, smoking is retained from the baseline, as the only significant variable. Using pooled simulations, veracity of the dataset is questionable but the results obtained in total cases appear to be very plausible.

to new components. The results indicated model sensitivity to the addition of new model components, at the granular level (looking at model coefficients and confidence or credible intervals).

The big picture indicated that the baseline SLR model was robust enough to withstand the addition of model components, evidenced by the retention of the same significant variables in the FPLR model. Moreover, despite model sensitivity at the granular level, the addition of model components led to improved estimation precision lending more credibility to the robustness of the significant coefficients reported in the published study.

0.4.2 Bayesian Logistic Regression (BLR)

The BLR model as a *global aleatory sensitivity analysis tool*, provided an opportunity to investigate the stability of the simple logistic regression (SLR) model. The BLR model was derived by adding a quadratic regularization term or component to the baseline log likelihood function of the SLR (Visit the Bayesian Logistic Regression part in the methods section).

The incorporation of a non-informative prior (zero mean normal distribution(s) on the model coefficients) into the SLR model, yields a quadratic penalty term similar to the familiar *Ridge* ($L2$) regularization component. It was clearly a hasty generalization, but the BLR results with the addition of an informative prior, were not expected to differ drastically from the BLR results with the non-informative prior; the sensitivity analysis did not bother with the implementation of 2 variants of the same model, on the assumption that the results would be the same. Regularization components are thought of as global sensitivity analysis tools because they have global effects on all coefficients.

The BLR model coefficients differed from the baseline SLR model, indicative of the SLR model sensitivity to new components. The results indicated model sensitivity at the granular level (looking at model coefficients and confidence or credible intervals) to the addition of new model components. The BLR model outputs, pointed towards local sensitivities in variable(s) like ***general health*** and ***income***. This is clearer in the ***income*** variable, where the coefficients of some levels that were observed to have negative effects ($OR > 1$) shift towards having positive effects ($OR < 1$); whereas these same coefficients straddled the threshold line in the FPLR model. Positively, there were no reversal in effects for the coefficients of significant variables; for example ***general health*** had negative effects ($OR > 1$) for all reported levels and the BLR model did not cause any reversal (Revisit the Bayesian Logistic Regression Results and Firth Penalized Logistic Regression Results to observe this phenomenon).

The big picture indicated that the baseline SLR model was robust enough to withstand the addition of model components, evidenced by the retention of the same significant variables in the BLR model. Moreover, despite model sensitivity at the granular level, the addition of model components led to improved estimation precision lending more credibility to the robustness of the significant coefficients reported in the published study.

0.4.3 Bootstrapping Simple Logistic Regression (SLR)

To make this accessible to every audience, bootstrapping asks a simple question, if the sample was NOT a close approximation of the population, how do point estimates and estimation precision differ? The randomness of the datasets in bootstrapping produced higher variance (in theory $\sim 38\%$ variance) between folds, making it excellent for simulating population diversity. However, bootstrapping excels in exposing any flaws in estimation precision (obtaining robust confidence intervals by virtue of the higher variance between folds), more so than pointing out any differences in the point estimates or coefficients.

Bootstrapping results showed that point estimates were sensitive to misrepresentation of the population, given the difference(s) in point estimates, relative to the baseline SLR model. Positively, there was no reversal in the effects of coefficients, meaning coefficients that had negative effects ($OR > 1$) retained those effects and vice-versa.

With bootstrapping, population(s) more diverse than represented by the sample, indicated poor estimation pre-

cision globally, but this was more pronounced in variable(s) like ***general health*** and ***income*** that showed high local sensitivities.

The implementation of bootstrapping inherently led to scenario(s) where the assumption of independence can be violated. The observations above automatically held true, for the violation of the assumption of independence.

The big picture indicated that the baseline SLR model outputs were robust enough to withstand poor representation of the population, evidenced by the retention of the same significant variables in the bootstrapped SLR model. Conversely, model sensitivity to potential population misrepresentation or violation of the independence assumption, indicated incredibility (unreliable estimation) of the model coefficients.

Bootstrapping showed that model point estimates were robust to random data configurations (misrepresentation of the population), but there were concerns in the realm of reliable estimation, hence casting some doubt on the robustness of the significant coefficients reported in the published study.

0.4.4 Cross Validating Simple Logistic Regression (SLR)

To make this accessible to every audience, leave one out cross validation (LOOCV) asks a simple question, if the sample was a close approximation of the population, how do point estimates and estimation precision differ? LOOCV excels in exposing any flaws in point estimation (obtaining robust point estimates by virtue of the extremely low variance between folds), more so than pointing out any differences in estimation precision.

LOOCV results showed that point estimates were insensitive to close approximation of the population, given the similarities in point estimates, relative to the baseline SLR model.

The big picture indicated that the baseline SLR model outputs were robust enough to withstand close approximation of the population by the used sample, evidenced by the retention of the same significant variables in the LOOCV SLR model. Moreover, LOOCV SLR produced the expected tight confidence intervals, lending more credibility (estimation reliability) to the robustness of the significant coefficients reported in the published study.

0.4.5 Simple Logistic Regression Sensitivity to Outliers

Outliers within samples can often be thought of high effect subjects that alter model outputs in ways that trigger model developers to reconsider running a model without these outliers.

Cook's distance was utilized as metric to determine influential data points that altered model coefficients. There was no clear characteristic about the subjects, marked as outliers by Cook's distance, except the measures of *deviance* and *leverage* utilized to calculate Cook's distance. Cook's distance formulation was outsourced to software because there was no clear concept on how to calculate Cook's distance for a logistic regression. However, a graph was utilized to find "truly" extreme outliers, which resulted in 4 subjects being classified as outliers.

The baseline SLR model was sensitive to outliers, evidenced by the global difference(s) in point estimates and confidence intervals in the SLR model without outliers. Baseline model output sensitivity to outliers was pronounced in the general health variable, indicated by large differences in the point estimates for some levels of the variable; and there were qualitative shifts indicated by the significance of the "General Health:Very Good", which was not significant in the baseline SLR model.

The big picture indicated that the baseline SLR model outputs were robust enough to withstand the presence of outliers, evidenced by the retention of the same significant variables in the SLR model without outliers. Moreover, SLR without outliers produced confidence intervals that appeared to have the same range as the baseline SLR model, lending more credibility (estimation reliability) to the robustness of the significant coefficients reported in the published study.

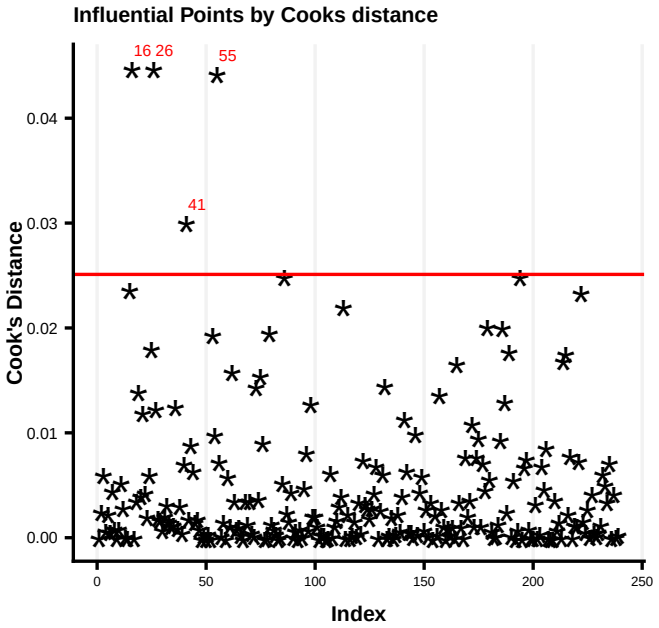


Figure 8: Graphical tool used to guide the excision of outliers using Cooks distance. The threshold (red line) was selected as 6/n, with the numerator selected on a trial and error basis in order to detect truly extreme outliers (n was the number of observations) . Cooks distance identified 4 alleged farmers as extreme outliers.

0.4.6 Knowledge Pertubations By Sample Size Variation

In statistics, models are often thought of as the brain responsible for processing knowledge which is "proxied" by data or the sample of interest. Knowledge often presents with 2 components, namely the quantity (amount) and quality. In statistics, the amount of knowledge is referred to as sample size, while the nature of quality is often determined by experimental design often with an emphasis on mitigating most or all risks from unmeasured confounding.

The performance of the SLR model was evaluated at different quantities of knowledge relative to the baseline quantity of knowledge (sample size is a determinant of how much knowledge the logistic regression model has to work with). Specifically, the SLR model was applied to sample of completed cases ($N_{cc} = 199 < N_{baseline} = 239$) and a sample of total cases ($N_{tc} = 2994 > N_{baseline} = 239$); essentially, evaluating scenarios were smaller vs larger amounts of knowledge were known about the sample space. Data for total cases was a simulated set, indicating that the veracity or quality of the information was questionable, although such datasets were expected to follow the same distributions observed in the data of known information.

For non-scientist, the assumption often is that more knowledge is often better no matter the problem at hand. In statistical science, the consensus often is "balanced" is always better, and of course, in some rare cases "more" or "less" might be better. The nature of the noise-signal spectrum, is one that naturally selects "balance", because "too much" can get ***noisy*** and "too little" can also get ***noisy***. The foundation of the "trade-off" or "balance" means that experimenting with different sample sizes, is no longer as predictable, as would be thought of by the non-scientist.

As expected, there was granular model sensitivity at different sample sizes that differed from the baseline sample size. Qualitatively, there was high local sensitivity around the general health variable reported, for completed cases ($n = 199$) when compared to the baseline ($n = 239$) or total cases ($n = 2994$.

The published or effective sample SLR was not robust to varying sample sizes, evidenced by the differences in significant variables obtained in completed cases vs total cases. In the completed cases SLR, 2 out of 3 variables considered significant in the effective sample SLR were retained as significant; ***smoking*** was no longer significant when completed cases were ingested by the SLR model. In the total cases SLR, 1 out of 3 variables, considered significant in the effective sample SLR were retained as significant; ***psychological distress*** and ***age*** were no longer significant, but ***race*** became a new significant variable.

The retention of ***smoking*** as significant in total cases was a fascinating finding and considered extremely plausible because ***smoking*** was thought of as a variable that could be more precisely measured than the main predictor ***psychological distress***. The caveat was, data on total cases was a simulated set and so the model outputs could only be valid if the veracity of information was ignored.

In the grand scheme, the baseline SLR model (effective sample model) was sensitive to variations in sample size. The general trend showed that global estimation precision improved with larger amounts of information (larger sample size), while deterioration of estimation precision was observed with smaller amounts of information (smaller sample size). Baseline SLR results were not considered robust because not all variables were retained at smaller or larger sample sizes; In high veracity (completed cases) ***smoking*** was not significant (came very close to being significant with $p = 0.055$ and even higher adjusted OR), with higher estimation uncertainty; In lower veracity (total cases simulated) ***smoking*** was significant, with lower estimation uncertainty. The baseline SLR model will need a larger sample size and "true" information to be considered robust enough.

0.4.7 Local Sensitivity Analysis on General Health

Suspicions of class imbalance within the general health variable, triggered arbitrary reclassification of the variable into a binary variable. The simple heuristic used assumed that, alleged farmers that reported "Excellent" or "Very Good" general health should be collapsed as "Good" general health, otherwise co"Poor" general health.

Comparing the newly classified general health SLR model to the baseline SLR model, would have been a classic case of comparing apples to oranges; hence the results were reported as a standalone model, aimed at investigate estimation precision and the significance of the general health variable.

The results indicated that collapsing this variable improved estimation precision. The significance of the variable was also retained making the original findings on general health robust towards this needed modification. The local sensitivity analysis had no unexpected global effects, indicating the baseline SLR model outputs were robust to local perturbations on a single variable.

0.4.8 Local Sensitivity Analysis on Psychological Distress

The original study provided a definition for ***psychological distress***. The ***psychological distress*** variable was considered confusing because there existed another variable that could be considered a ***psychological distress*** variable.

To investigate unmeasured confounding around ***psy-***

chological distress, the ***number of bad mental health days*** was used to investigate the original definition of psychological distress. Alleged farmers classified as psychologically distressed, that reported greater than zero days (> 0) of bad mental health, were confirmed as psychologically distressed; otherwise, they were NOT psychologically distressed. There were considerations of using a threshold greater than 0 for the confirmation of ***psychological distress***, given that previous research utilizing *Principal Component Analysis (PCA)* had provided a hint of ~ 9 bad mental health days, on the 30 day scale [adeck2026mental]. Nonetheless, any value greater than 0 seemed like a good starting point for determining the number of bad mental health days, needed to confirm psychological distress. There expectation was NO difference in the significance of ***psychological distress*** after redefining the variable, because the expectation was the number of bad mental health days greater than 0 should be complementary to the original definition of psychological distress (Visit the Appendix for a description of this variable).

Accounting for unmeasured confounding around ***psychological distress***, rendered the variable statistically insignificant. The baseline SLR model output for ***psychological distress***, was not robust enough to withstand unmeasured confounding around the variable. The local sensitivity analysis had no unexpected global effects, indicating the ***other*** baseline SLR model outputs were robust to local perturbations on a single variable (***smoking***, ***age*** and ***general health*** retained their statistical significance).

Did the authors misconstrue social distress as psychological distress? The survey question utilized by the authors, did not seem to provide clarity as to what term should be used to describe the variable. There seemed to be some unmeasured confounding in the measure of psychological distress either from survey design, subject misunderstanding of the question, misinterpretation of the variable by the authors, complication(s) from the sensitivity analysis; irrespective of the confounding source, psychological distress was sensitive to unmeasured confounding and became insignificant when redefined.

0.4.9 Synthesis

True perturbations of knowledge through simulation indicate that ***smoking*** might be the causal factor linked to ***COPD***; the causal claim is bold because it is difficult to imagine a world where a ***COPD*** subject will turn to ***smoking*** as a cure. ***Smoking*** was insignificant (by a very small margin $p = 0.055$) when the SLR model was applied to completed cases, but this was likely due to losing useful data on smoking, which was believed would have rendered it significant. Moreover, ***smoking*** appeared to be the variable that could be measured with the highest precision, making the results from the simulated dataset highly plausible.

Additionally, unmeasured confounding around the variable dubbed ***psychological distress*** by the authors, wiped out the significance of the variable, further solidifying the claim that ***smoking*** might be the causal factor.

All results indicated that the authors had plausible results tied to the sample they used. Any true deviations from the published sample, like simulations for total cases or using completed cases, had the potential to challenge the core findings (significant variables) from the published paper.

0.5 Limitations

The limitations are discussed in the subsections that follow.

0.5.1 Simulations

The SLR model on total cases, utilized simulated data. Discussions revealed that utilizing the larger sample size in total cases, led to improved estimation precision and perhaps retained the true causal factor (***smoking***). With no "true" larger sample, the results from the baseline SLR outputs are considered robust enough; although there is a high chance the model results might not be robust enough.

0.5.2 Resampling

Bootstrapping or LOOCV provide means to understand how model coefficients behave in scenarios of remote approximate of the population and close approximation of the population, respectively. SLR model outputs from bootstrapping or LOOCV, helps in answering questions more about the used sample and how it might translate or be extrapolated towards the population, but it fails to account for the "real" population. The concerns with hypothetical populations generated by Bootstrap or LOOCV, can be mitigated by applying the SLR model to "real" data from other years (2019 etc), in order to confirm the model outputs and their robustness.

0.5.3 Local Sensitivity Analysis

In local sensitivity analysis, heuristics were used in recoding or redefining variables. With no domain knowledge or literature reference for the local changes to variables, the modified SLR model(s) might be unreliable in determining the robustness of model coefficients from the baseline SLR.

0.6 Conclusions

The *Bayesian logistic regression (BLR)* model, indicated the presence of global ***aleatory uncertainty*** in the *simple logistic regression (SLR)* model outputs; global ***aleatory uncertainty***, was ascertained by the reversal in the effects of the coefficients (revisit the discussion on

Bayesian Logistic Regression), for certain levels of the ***income*** variable, albeit this variable being statistically insignificant. Global sensitivity of the ***income variable*** to additional model components did not provide enough evidence to dispute the robustness of the core findings (significant variables) by Kham-Ai, Heaton, and Li (2023), that ***psychological distress***, ***smoking***, ***age*** and ***general health*** were significant variables in determining the odds of having ***COPD***. Conclusively, there was ***global aleatory uncertainty*** in the *simple logistic regression (SLR)* model outputs, but not relevant in disputing the core findings (significant variables) from the published paper by Kham-Ai, Heaton, and Li (2023).

Global ***epistemic uncertainty*** in *simple logistic regression (SLR)* model outputs driven by data was a de facto expectation, given no experimental design is perfect. All investigations for global ***epistemic uncertainty***, indicated its presence. Utilization of simulation for total cases ($N = 2994$) produced fascinating results, which challenged the core findings (significant variables) from the published study by Kham-Ai, Heaton, and Li (2023); the ***smoking*** variable appeared to be the only significant variable retained from the core findings (significant variables), when the simulated datasets of total cases were used in the *simple logistic regression (SLR)* model. If the simulation of total cases was close to an approximation of the "truth", then the core findings (significant variables) from the original study by Kham-Ai, Heaton, and Li (2023) were not robust enough to withstand the incorporation of total knowledge (total cases). Conclusively, the use of simulation likely failed the veracity criteria, rendering the original core findings (significant variables) from Kham-Ai, Heaton, and Li (2023), robust enough.

Local ***epistemic*** sensitivity analysis on ***general health***, led to the conclusion that its coefficient could be reliably estimated if the levels within the variable were collapsed, in a way that facilitated interpretation. Conclusively, retention of statistical significance of the ***reclassified general health*** variable meant that the original core findings (significant variables) from Kham-Ai, Heaton, and Li (2023), were robust enough.

Local ***epistemic*** sensitivity analysis on ***psychological distress***, led to the conclusion that the coefficient was statistically insignificant, if unmeasured confounding in the variable was accounted for. Conclusively, the original authors (Kham-Ai, Heaton, and Li (2023)) needed to reevaluate their definition of ***psychological distress*** and consider the possibility that ***social distress*** might have been the main predictor, but they reported it as ***psychological distress***.

Conclusively, all investigations tied to the published sample (Kham-Ai, Heaton, and Li (2023)) indicated that the core findings (significant variables) from model outputs were robust enough towards uncertainty (aleatory or epistemic), if the published sample size was sufficient and unmeasured confounding around psychological distress was ignored.

0.7 Future Work

The published paper focused on understanding how psychological distress and covariates affected the odds of having ***COPD*** (the authors were focused on hypothesis testing). Performing a ***Power Sensitivity Analysis*** is a next logical step, in determining if the sample size (given the obtained coefficients) was large enough to provide $> 80\%$ power, in rejecting the null hypothesis.

Incorporation of Air Quality (AQ) information by state [1], into the published sample could be insightful in understanding how air quality affects the odds of ***COPD***. Specifically, numeric ranking of air quality for states could be used in a cumbersome mixed effects model, to understand how air quality rank affects the ***COPD*** interaction with psychological distress and covariates.

Bootstrapping and LOOCV provided resampling tools for simulating a population using actual data. The alternate to simulation will be applying the SLR model to 2019 BRFSS data that has the same architecture or variables like the 2017 BRFSS data. Previous research by Adeck (2026) [2] indicated that the published research by Kham-Ai, Heaton, and Li (2023) will not be possible with 2024 BRFSS data.

Application of ***machine learning*** models like ***Gradient Boosting***, ***Classification and Regression trees (CART)*** to rank the most important variables can help in changing the narrative from simple interaction between ***COPD*** and ***psychological distress*** and ***smoking***, to determining a causal relationship. Application of ***Principal Component Analysis (PCA)*** to the published sample can help in detecting the main clusters within the data, further helping in making causal claims, as opposed to interaction claims.

A new SLR model that defines ***psychological distress*** using the ***number of bad mental health days***, either as a numeric variable or converted into a binary factor, while keeping the same covariates. This will be done to address the concerns surrounding the definition of ***psychological distress***.

A nominal regression model with ***COPD severity*** as the outcome variable, while maintaining the same main predictor and covariates in the model by Kham-Ai, Heaton, and Li (2023). ***COPD severity*** is a good measure for staging ***COPD***, but the variable is not presented as a dichotomous variable because it has more than 2 levels. Hence, a nominal regression approach will be needed to understand what factors drive ***COPD severity***.

[1] Visit the World Population Review page, or National Institute on Minority Health and Health Disparities for more information.

[2] The paper by Adeck (2026) will be attached given it is an unpublished source. Otherwise, feel free to request the paper from the author.

0.8 *References*

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0.9 Appendix

0.9.1 Variables or Measures Continuation

COPD: An individual was classified as having COPD symptoms, if they answered "yes" to one of three questions indicating in the past 3 months if they had a cough on most days, coughed up phlegm or mucus on most days, and/or had shortness of breath either when hurrying on level ground or when walking up a slight hill or stairs. This information is provided for replication of the sensitivity analysis (Kham-Ai, Heaton, and Li 2023).

Age (_AGE80): Age was a measure of length of living in years.

Race (_RACE): Race was a calculated variable that was re-coded into a binary variable to represent "White" and any race not white dubbed "Others".

Gender (SEX): Farmer gender.

Redefined Psychological Distress (SDHSTRES_SA): The redefined psychological distress variable classified psychological distressed farmers that reported greater than 0 days of poor mental health, as psychologically distressed, otherwise, not in psychological distress. The number of poor mental health days was obtained from the query, "Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?".

General Health (GENHLTH): The general health variable for farmers, was obtained from answers to the query "Would you say that in general your health is?". Valid responses were, "Excellent", "Very Good", "Good", "Fair", and "Poor".

Reclassified or Collapsed General Health (GENHLTH_SA): Farmers that reported general health as "Excellent", "Very Good", or "Good" were classified as 'Good" health, while general health reported as "fair" or "poor" were classified as "Poor" health.

Smoking Status (SMOKE100): The variable was derived from a response to "Have you smoked at least 100 cigarettes in your entire life: [Note: 5 packs = 100 cigarettes]? Valid responses were"Yes" or "No".

Income (_INCOMG): The variable was a calculated or computed variable from the query "Is your annual household income from all sources: (If respondent refuses at any income level, code ´Refused.´)?". The computed variable produced 5 valid categories.